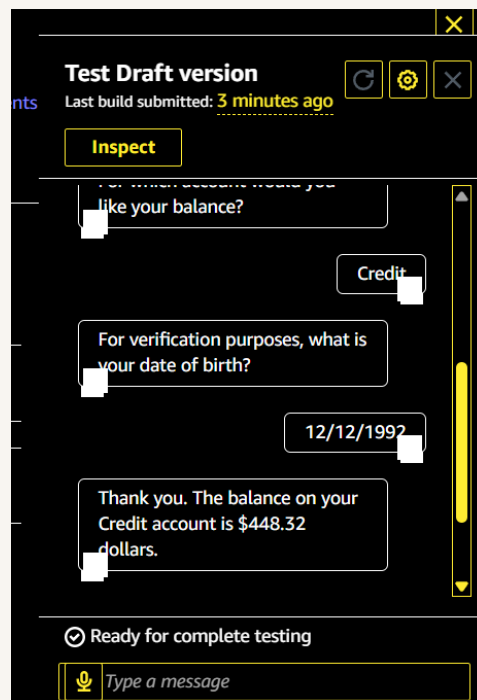




Connect Amazon Lex with Lambda



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Introducing Today's Project!

What is Amazon Lex?

Amazon Lex is a fully managed artificial intelligence (AI) service from AWS that lets you build voice and text conversational interfaces using Natural language.

How I used Amazon Lex in this project

In today's project, I used Amazon Lex to connect my Bot to Lambda in order to complete an action that it cannot alone (Giving random figures to users when they ask after their account Balance).

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how much time I had to spend writing and understanding the Lambda function.

This project took me...

This project took me 2 hours



AWS Lambda Functions

AWS Lambda is a service that lets me run code in the cloud without needing to manage any computers/servers - Lambda will manage them for me. Lambda runs the code only when needed and scales automatically, from a few requests per day to thousands per second - all i need to do is supply my code in one of the languages that Lambda supports.

In this project, when someone asks about their account balance to my chatbot, Lex will ask my Lambda function to run this code (The BankLambdaFunction.py), which will pick a random number to pretend it's the balance Lambda will pass this random number to Lex, who will then push the bank balance figure to the user through my chatbot.



```
BankLambdaFunction.py 9+ X
C:\Users\m2ja>OneDrive>Documents>Cloudingineer>AWS_Projects>My_Projects>AWS_AI_Services>Amazon_Lex>Part 3>BankLambdaFunction.py
1 """
2 How does AWS Lambda cheer up Amazon Lex? By saying, "Don't worry, I've got your back back(end)!"
3
4 """
5
6 import json
7 import random
8 import decimal
9
10
11 def random_num():
12     return decimal.Decimal(random.randrange(1000, 50000)) / 100
13
14
15 def get_slots(intent_request):
16     return intent_request["sessionState"]["intent"]["slots"]
17
18
19 def get_slot(intent_request, slotName):
20     slots = get_slots(intent_request)
21     if slots is not None and slotName in slots and slots[slotName] is not None:
22         return slots[slotName]["value"]["interpretedValue"]
23     else:
24         return None
25
26
27 def get_session_attributes(intent_request):
28     sessionState = intent_request["sessionState"]
29     if "sessionAttributes" in sessionState:
30         return sessionState["sessionAttributes"]
31     return {}
32
33
34
35 def elicit_intent(intent_request, session_attributes, message):
36     return {
37         "sessionState": {
38             "dialogAction": {"type": "ElicitIntent"},
39             "sessionAttributes": session_attributes,
40         },
41         "messages": [message] if message != None else None,
42         "requestAttributes": intent_request["requestAttributes"]
43         if "requestAttributes" in intent_request
44         else None,
45     }
```



Chatbot Alias

An alias is a pointer for a specific version of a bot. So when connecting Lex with other AWS services or custom applications, those external resources will connect to an alias, which will point to the specific version of the bot that i want to use

TestBotAlias is a default version of a bot that's made for testing or development. It can be used to make sure everything works smoothly before rolling out changes!

To connect Lambda with my BankerBot, I visited my bot's TestBotAlias and Associate my bot with my Lambda function, leaving the function version at the default \$LATEST. This version enables me to direct my alias to always use the most up to date version of this Lambda function.

Alias language support: English (US)

▼ **Lambda function - optional**
This Lambda function is invoked for initialization, validation, and fulfillment.

Source
lambda_function ▼

Lambda function version or alias
\$LATEST ▼

[Learn more about Lambda](#)

Cancel Save

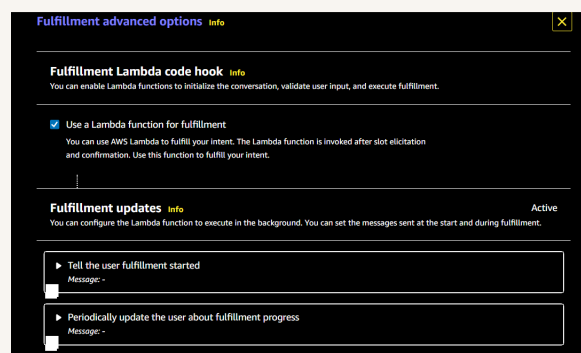


Code Hooks

A code hook helps connect the chatbot to custom Lambda functions for doing specific tasks during a conversation. They're used to handle more complex actions that the basic chatbot setup can't do on its own, like checking data from a database or making decisions based on past conversations.

Even though I already connected my Lambda function with my chatbot's alias, I had to use code hooks because even though the lambda function is connected it is not active. Code hook in the corresponding intent provide an option to activate it.

I could find code hooks at the fulfillment panel of the corresponding intent, (Advanced Options), and checked the box to activate the setting to Use a Lambda function for fulfillment.





The final result!

I've set up my chatbot to trigger Lambda and return a random dollar figure when the user has choosed his account type and the verification with the date of birth has been completed.

